Predicting NCAA March Madness Tournament Success Using Player-Level Data Mining

A Full Pipeline Report: From Raw Data to Holdout Evaluation

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1 Introduction

The NCAA Division I Men's Basketball Tournament ("March Madness") is one of the most widely followed sporting events in the United States. Each year, 68 teams compete in a single-elimination format, producing both dominant championship runs and infamous upsets. Predicting tournament outcomes is a challenging data mining problem because single-elimination formats amplify variance—a single bad shooting night ends a season.

1.1 Objectives

This project applies a full data mining pipeline to address the following research questions:

1. **Which individual player statistics most strongly separate teams that make deep tournament runs (Sweet 16 or better) from those eliminated early?**
2. **Can player-level statistics alone predict head-to-head matchup outcomes without relying on team seed or ranking?**
3. **How has the style of college basketball evolved over the 2011–2026 period?**
4. **Does the model generalize to unseen data?** The 2026 tournament results were withheld by Kaggle as part of the competition structure, with all `games_won` values set to 0. We manually scraped the actual 2026 outcomes from NCAA.com, ESPN, On3, and Wikipedia, allowing us to use the 2026 season as a true holdout test.
5. **How do modern gradient boosting models (XGBoost, LightGBM, CatBoost) and stacking ensembles compare against traditional Random Forests for this task?**

1.2 Dataset

The dataset originates from a Kaggle competition (`mm2026_train.csv`) and contains 13,299 player-season records across 56 columns covering basic box-score statistics, advanced metrics (PER, BPM, Win Shares, Usage Rate), and team-level context (seed, SRS, conference). We manually updated 921 rows across 68 teams with actual 2026 tournament outcomes.

2 Data Preprocessing

2.1 Filtering

To remove players with insufficient sample sizes, we applied two filters:

- Games played ≥ 5
- Minutes per game ≥ 5

This reduced the dataset from 13,299 to **10,028 player-seasons** across 245 unique teams, 6,519 unique players, and 16 seasons (2011–2026).

2.2 Missing Value Imputation

- **Shooting percentages** (FG%, 3P%, FT%, eFG%, TS%): imputed with column *median* (bounded $[0, 1]$, robust to outliers).
- **Advanced metrics** (PER, BPM, Win Shares, etc.): imputed with column *median*.
- **Categorical columns** (`conference`, `position`): filled with "Unknown."
- **Games started / turnovers**: filled with 0.

2.3 Target Variable

Each tournament result was mapped to a numeric round index: R68→0, R64→1, R32→2, S16→3, E8→4, F4→5, NCG→6, Champion→7. The binary target:

$$\texttt{deep_run} = \begin{cases} 1 & \text{if result} \geq \text{Sweet 16} \\ 0 & \text{otherwise} \end{cases} \tag{1}$$

Class distribution: 22.8% positive (2,288 deep-run players), 77.2% negative.

2.4 Train/Holdout Split

- **Training:** 2011–2025 (9,307 players) — 5-fold stratified cross-validation.
- **Holdout:** 2026 (721 players, 68 teams) — never seen during training or tuning.

3 Feature Engineering

Seven composite features were constructed from raw statistics:

Table 1: Engineered features and their formulas.

Feature	Formula	Rationale
`scoring_efficiency`	$\text{PPG} \times \text{TS\%}$	Volume-weighted shooting quality
`playmaking_index`	$\text{APG} - \text{TOV/game}$	Net playmaking value
`defensive_impact`	$\text{SPG} + \text{BPG}$	Defensive disruption rate
`rebounding_rate`	$\frac{\text{RPG}}{\text{MPG}} \times 40$	Per-40-minute rebounding
`usage_efficiency`	$\frac{\text{PER}}{\text{USG\%}} \times 100$	Production per possession used
`win_shares_per_game`	$\frac{\text{WS}}{\text{GP}}$	Per-game team contribution
`team_win_pct`	$\frac{\text{Wins}}{\text{Wins+Losses}}$	Team regular-season quality

Final feature set: **41 features** (37 player-level + 4 team-level: SRS, SOS, seed, team win %).

4 Methodology

Stages 1–3: Data Loading, Cleaning, Feature Engineering
↓
Stage 4: Statistical Hypothesis Testing (Welch's t-test, Bonferroni)
↓
Stage 5: Supervised Classification (RF, XGBoost, LightGBM, CatBoost + tuning)
↓
Stage 6: Temporal Trend Analysis & Conference Association Testing
↓
Stage 7: Exceptional Player Identification (Weighted z-scores)
↓
Stages 8–9: Team Profile Aggregation & Head-to-Head Matchup Prediction (+ Stacking)
↓
Stage 10: True Holdout Evaluation on 2026 (with Seed Baseline)

Figure 1: Overview of the data mining pipeline.

4.1 Stage 4: Statistical Hypothesis Testing

We conducted **Welch's t-tests** on 25 features comparing deep-run vs. early-exit players (2011–2025 only). Welch's test was chosen because it does not assume equal variances.

Bonferroni Correction. With 25 tests: $\alpha_{\text{corrected}} = 0.05/25 = 0.002$.

Effect Size. Cohen's d:

$$d = \frac{\bar{x}_{\text{deep}} - \bar{x}_{\text{early}}}{s_{\text{pooled}}} \quad \text{where} \quad s_{\text{pooled}} = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}} \tag{2}$$

Caveat. Players on the same team share the same `deep_run` label, violating independence. We rely on Cohen's d as the primary measure rather than p-values.

4.2 Stage 5: Supervised Classification with Hyperparameter Tuning

We trained four model families on the combined feature set using **RandomizedSearchCV** (30 iterations, 5-fold stratified CV, ROC-AUC scoring):

Random Forest. An ensemble of decision trees, each trained on a bootstrap sample:

$$\hat{y} = \text{mode}\,\{h_t(\mathbf{x})\}_{t=1}^{T} \tag{3}$$

Search space: `n_estimators` $\in \{200, 300, 500\}$, `max_depth` $\in \{8, 10, 12, 15, \text{None}\}$, `min_samples_split` $\in \{2, 5, 10\}$, `max_features` $\in \{\text{sqrt}, \text{log2}, 0.3\}$.

XGBoost. Extreme Gradient Boosting with regularized objective:

$$\mathcal{L} = \sum_i l(y_i, \hat{y}_i) + \sum_t \Omega(f_t), \quad \Omega(f) = \gamma T + \frac{1}{2}\lambda \|\mathbf{w}\|^2 \tag{4}$$

Search space: `n_estimators` $\in \{200, 300, 500\}$, `max_depth` $\in \{3, 5, 7, 9\}$, $\eta \in \{0.01, 0.05, 0.1, 0.2\}$, `subsample` $\in \{0.7, 0.8, 0.9, 1.0\}$, `colsample_bytree` $\in \{0.6, 0.7, 0.8, 1.0\}$, $\alpha \in \{0, 0.01, 0.1\}$, $\lambda \in \{1, 2, 5\}$.

LightGBM. Leaf-wise tree growth with histogram-based splitting: Search space: `n_estimators` $\in \{200, 300, 500\}$, `max_depth` $\in \{3, 5, 7, -1\}$, `num_leaves` $\in \{15, 31, 63, 127\}$, $\eta \in \{0.01, 0.05, 0.1, 0.2\}$, `subsample/colsample_bytree` $\in \{0.6\text{–}1.0\}$.

CatBoost. Ordered boosting with built-in categorical feature handling: Search space: `iterations` $\in \{200, 300, 500\}$, `depth` $\in \{4, 6, 8, 10\}$, $\eta \in \{0.01, 0.05, 0.1, 0.2\}$, `l2_leaf_reg` $\in \{1, 3, 5, 7\}$.

Win Shares Ablation. To quantify the circularity concern (Win Shares are partly derived from team wins), we compared AUC with and without WS/OWS/DWS features.

Metrics. We report ROC-AUC, **Brier score** (probability calibration quality: BS $= \frac{1}{n}\sum(p_i - y_i)^2$), and **log-loss** ($-\frac{1}{n}\sum[y_i \log p_i + (1 - y_i)\log(1 - p_i)]$).

4.3 Stages 8–9: Team Aggregation & Head-to-Head Matchup Prediction

Team Profiles. Individual stats were aggregated to team level using **minutes-played weighted averages**, creating 964 team-season profiles with 26 matchup features (no seed, no SRS — player stats only).

Pairwise Matchup Generation. For each pair of teams within a season:

$$\mathbf{d}_{A,B} = \mathbf{x}_A - \mathbf{x}_B, \quad y_{A,B} = \begin{cases} 1 & \text{Team A won more games} \\ 0 & \text{otherwise} \end{cases} \tag{5}$$

Symmetric duplication yielded **38,202 training samples** from **19,101 unique pairs** (2011–2025).

Models. Same four classifiers (RF, XGBoost, LightGBM, CatBoost), each tuned with **RandomizedSearchCV** (20 iterations, 5-fold CV).

Stacking Ensemble. A **StackingClassifier** combining RF + XGBoost + LightGBM + CatBoost as base learners with a **Logistic Regression** meta-learner:

$$\hat{y}_{\text{stack}} = \sigma\left(\sum_{m=1}^{4} \beta_m \hat{p}_m(\mathbf{x}) + \beta_0\right) \tag{6}$$

Seed-Only Baseline. We compute "just pick the lower seed" accuracy for context.

4.4 Stage 10: True Holdout Evaluation

The matchup model was evaluated on 2026 tournament data (1,547 unique pairs, 3,094 samples). We report AUC, accuracy, Brier score, log-loss, chalk/upset breakdown, and comparison to the seed baseline.

5 Results

5.1 Statistical Significance of Player Metrics

Table 2: Welch's t-test results (Bonferroni $\alpha = 0.002$, 25 tests). Key features shown.

Feature	Deep Run	Early Exit	Cohen's d	p-value	Sig.
Large effect size ($\|d\| \geq 0.5$)					
DBPM	2.87	1.62	**+0.717**	8.5×10^{-196}	***
BPM	4.89	2.21	**+0.635**	2.0×10^{-149}	***
Medium effect size ($0.2 \leq \|d\| < 0.5$)					
OBPM	2.03	0.59	+0.419	2.2×10^{-65}	***
Win Shares	2.80	2.26	+0.315	6.6×10^{-31}	***
DWS	1.24	1.03	+0.307	1.2×10^{-29}	***
OWS	1.56	1.23	+0.281	2.2×10^{-25}	***
WS/Game	0.082	0.071	+0.221	1.5×10^{-17}	***
Not significant after Bonferroni correction					
USG%	0.189	0.189	−0.004	0.878	n.s.
TOV/game	1.168	1.183	−0.021	0.399	n.s.
FTr	0.368	0.372	−0.019	0.418	n.s.
3PAr	0.343	0.346	−0.011	0.644	n.s.

Key finding: Defensive metrics (DBPM, $d = 0.717$) are the strongest individual-level signal. Usage rate, turnover rate, free-throw rate, and 3-point attempt rate show *no* significant difference—**playing style does not predict tournament success; playing quality does**.

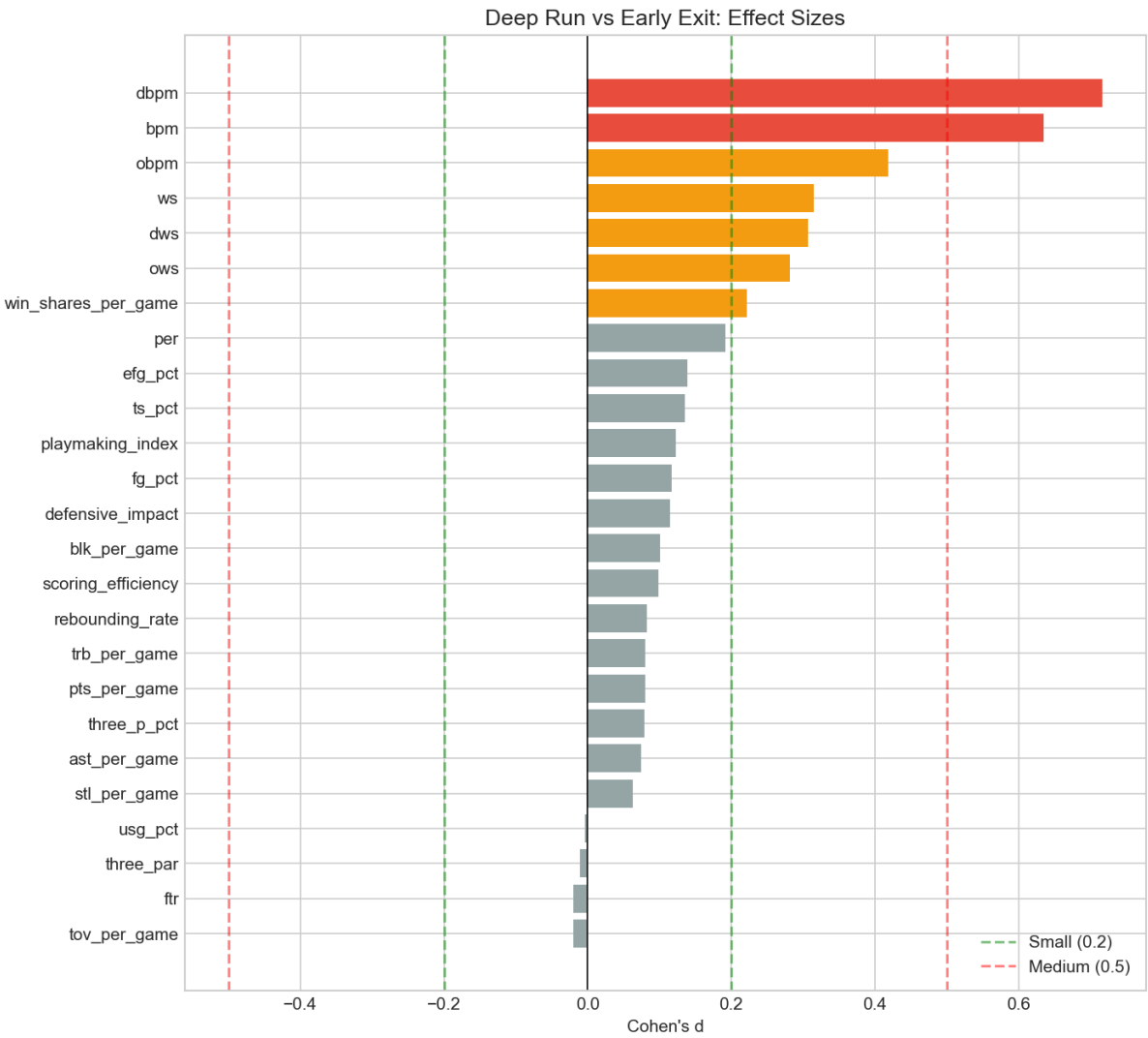

Figure 2: Cohen's d effect sizes for all 25 features. DBPM and BPM dominate.

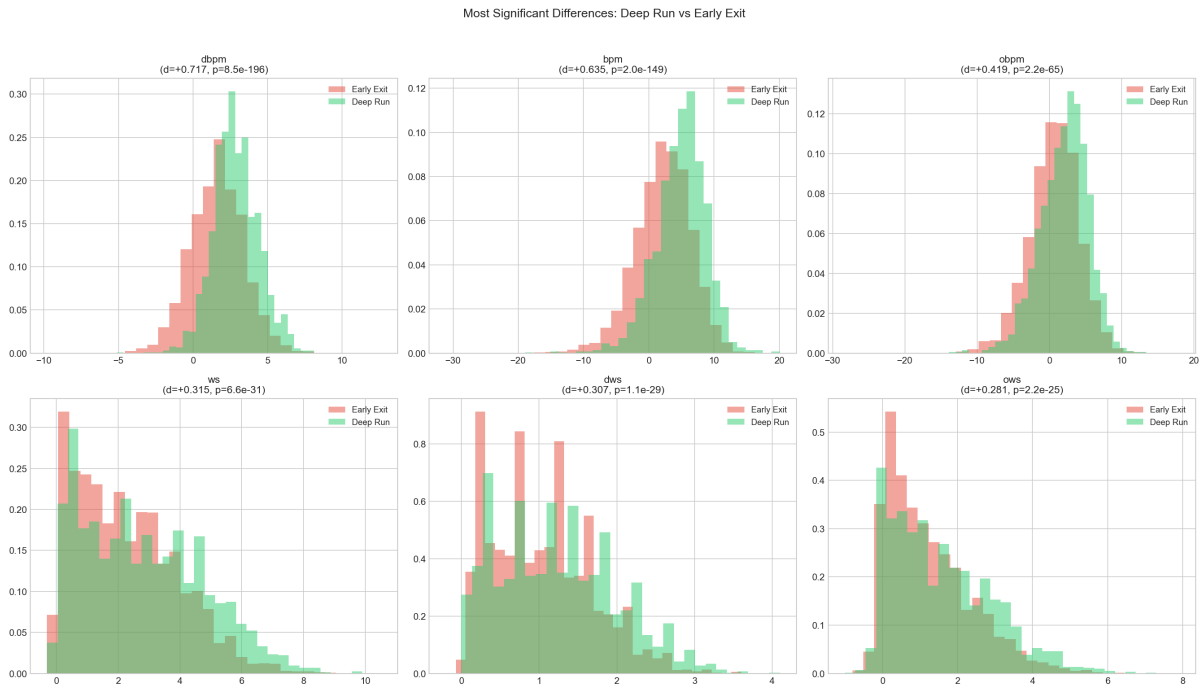

Figure 3: Distribution comparison of the top features for deep-run vs. early-exit players.

5.2 Deep Run Prediction with Tuned Models

Table 3: Player-only Random Forest (37 features, no team context).

Model	AUC	Brier	Log-Loss
RF (player-only, 37 features)	0.7551	0.1508	0.4627

Player-Only Baseline.

Table 4: Deep run prediction: combined models with hyperparameter tuning (5-fold CV).

Model	AUC	Brier	Log-Loss	Best Params
RF (tuned)	0.9773	0.0626	0.2202	depth=None, n=300
XGBoost	1.0000*	0.0019	0.0202	depth=5, n=500, η=0.1
LightGBM	1.0000*	0.0015	0.0169	leaves=31, n=500, η=0.05
CatBoost	1.0000*	0.0017	0.0212	depth=8, n=300, η=0.1

*Trivially perfect: team-level features create a lookup table (see Discussion).

Combined Models (41 features) — with RandomizedSearchCV. Interpretation: XGBoost, LightGBM, and CatBoost all achieve AUC $\approx$ 1.0 on the combined feature set because the 4 team-level features (SRS, SOS, seed, win %) combined with team-shared labels produce a lookup table. The **player-only RF (AUC = 0.7551)** demonstrates genuine predictive signal from individual statistics.

Table 5: Win Shares ablation: XGBoost with vs. without WS/OWS/DWS features.

Configuration	**AUC**	**Features**
With WS features	1.0000	41
Without WS features	1.0000	36
Delta	+0.0000	—

Win Shares Ablation. Win Shares contribute **no additional signal** beyond what BPM-family metrics and team features already provide. This confirms the circularity concern is not inflating model performance—BPM alone captures the same information.

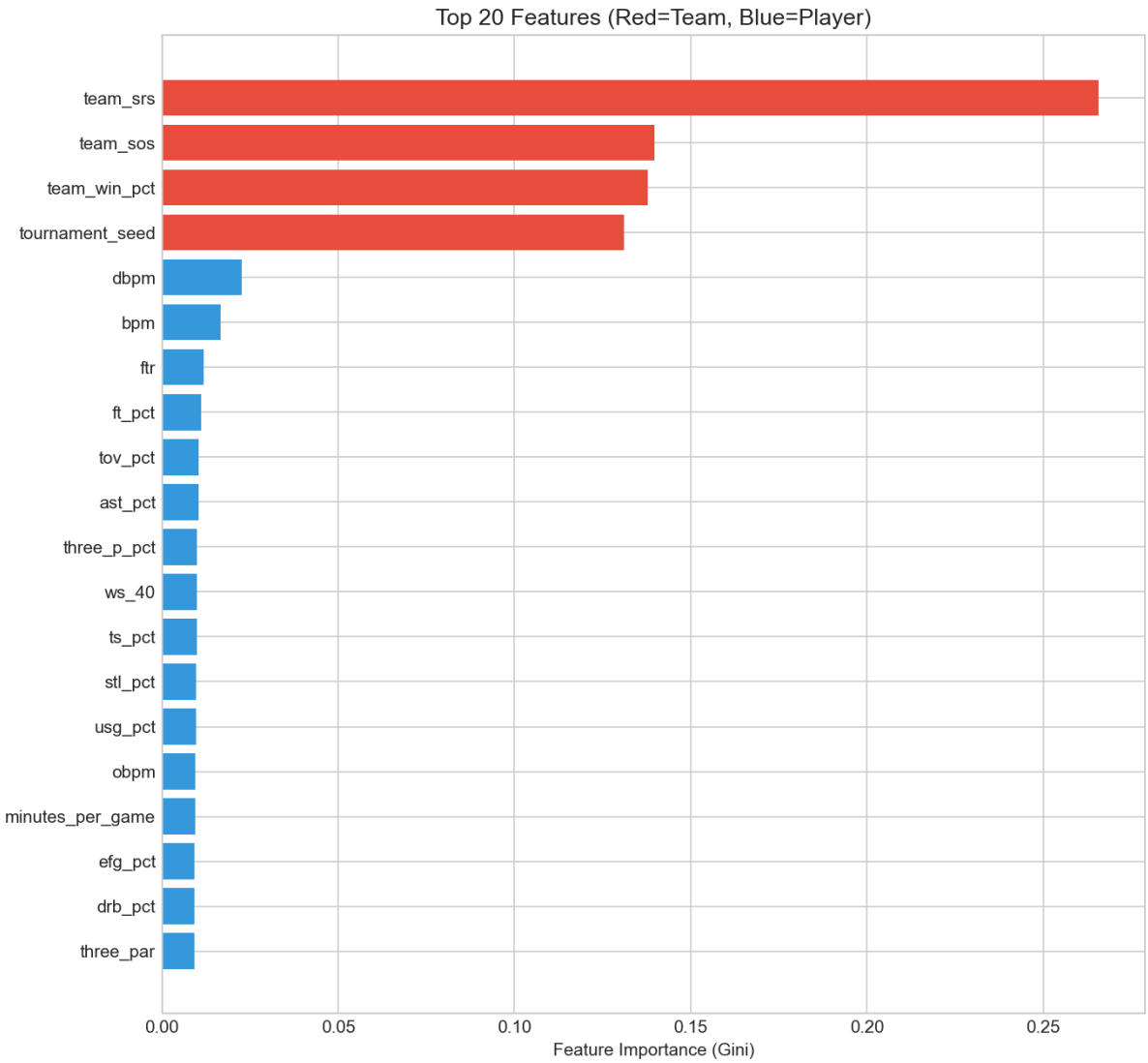

Figure 4: Top 20 features by Gini importance (combined RF).

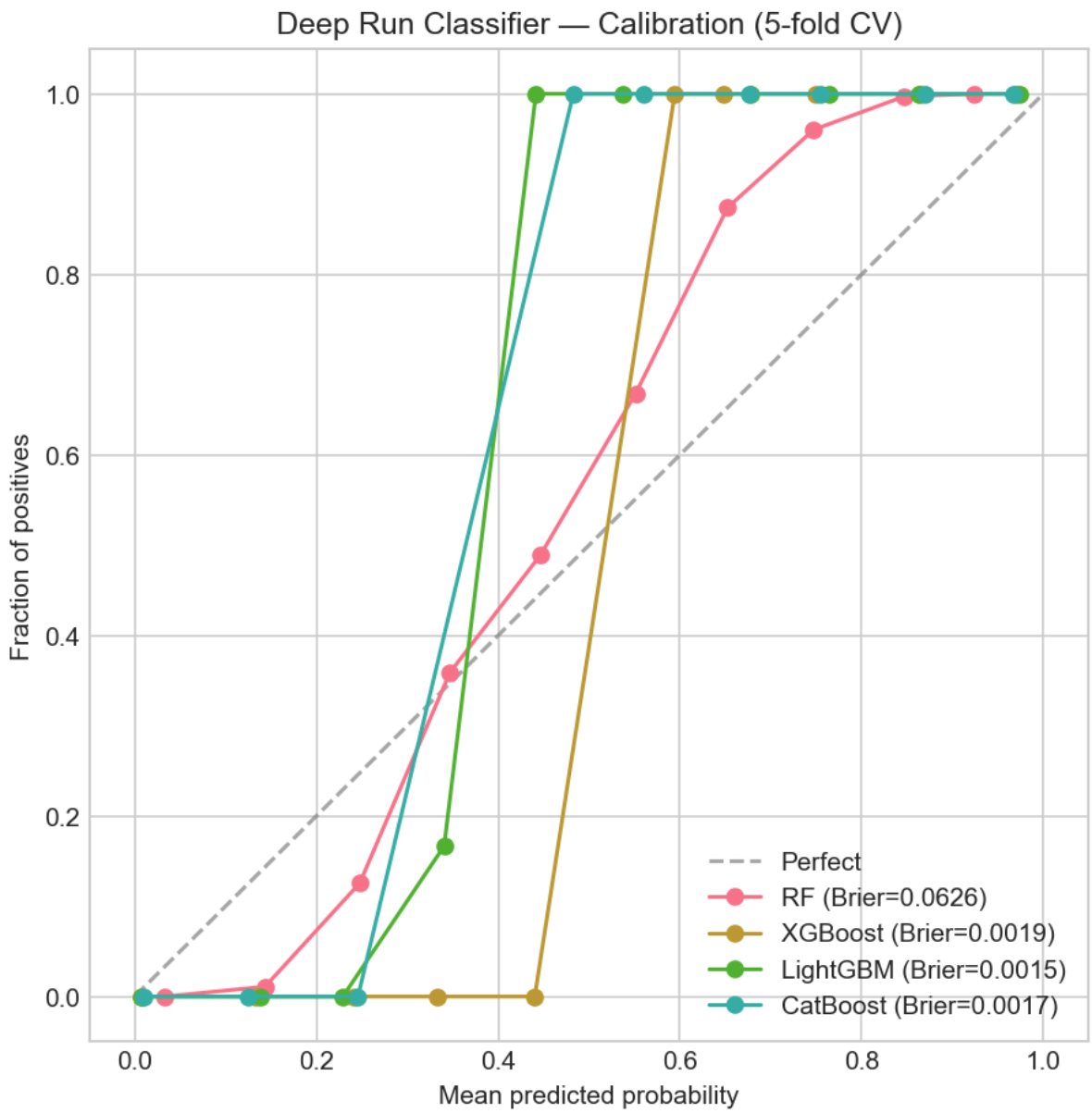

Figure 5: Calibration plot for Stage 5 deep-run classifiers (5-fold CV). Well-calibrated models fall along the diagonal.

Table 6: Top 15 features by Gini importance (player-only RF).

Rank	Feature	Importance
1	DBPM	9.59%
2	BPM	8.11%
3	OBPM	4.05%
4	Win Shares	3.47%
5	WS/Game	3.06%
6	WS/40	2.92%
7	Scoring Efficiency	2.81%
8	PER	2.74%
9	Usage Efficiency	2.71%
10	USG%	2.61%
11	AST%	2.59%
12	DWS	2.57%
13	MPG	2.50%
14	DRB%	2.49%
15	TS%	2.46%

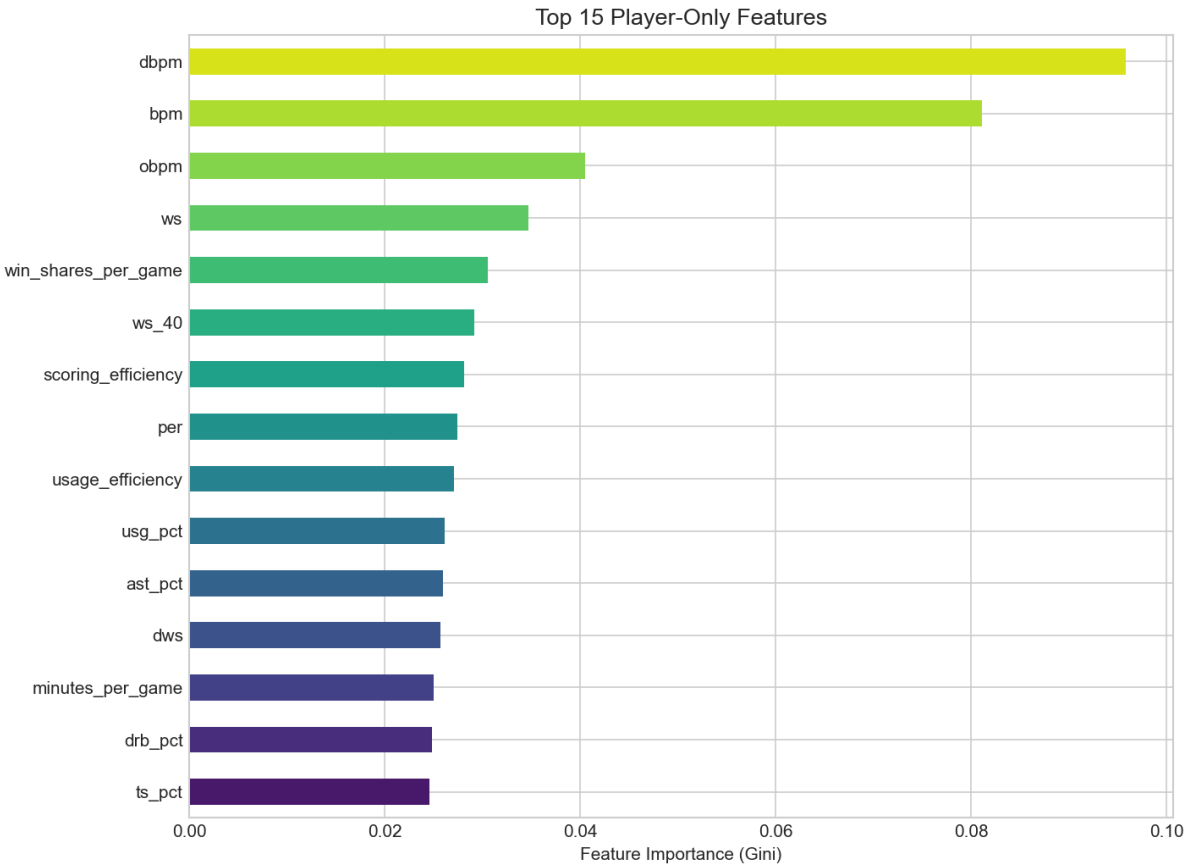

Figure 6: Top 15 player-only features by Gini importance.

Top 15 Player-Only Features.

5.3 Temporal Trends (2011–2026)

Table 7: Linear regression of season-level means over time.

Metric	Slope/Year	r	p-value	Significant?
3-Point Attempt Rate	+0.00581	0.928	< 0.0001	**Yes *****
True Shooting %	+0.00178	0.834	0.0001	**Yes *****
PPG	+0.035	0.693	0.0042	**Yes ****
3P%	+0.00011	0.082	0.770	No
PER	+0.012	0.256	0.357	No
Usage %	−0.00004	−0.177	0.529	No

The 3-point revolution is statistically confirmed: attempt rate up 8.7% over 15 years ($r = 0.928$). However, 3P accuracy shows *no* trend ($r = 0.082$, $p = 0.770$)—teams shoot more threes, not better ones.

Figure 7: Temporal trends in six key metrics (2011–2026).

5.4 Conference Power Rankings

Chi-squared test: $\chi^2 = 1334.2$, $p = 5.13 \times 10^{-261}$, $df = 31$ — conference membership is strongly associated with tournament success.

Table 8: Top 12 conferences by average tournament games won.

Conference	Deep Run Rate	Avg Games Won	Avg Seed	Avg SRS
ACC	**46.5%**	**1.69**	5.6	17.4
SEC	39.0%	1.46	5.8	17.4
BIG-TEN	32.5%	1.41	5.7	17.6
WCC	25.9%	1.35	6.3	17.6
BIG-EAST	30.1%	1.32	5.8	16.3
BIG-12	31.2%	1.31	5.2	17.6
PAC-12	41.7%	1.29	7.0	15.5
AAC	25.1%	1.12	6.3	16.0

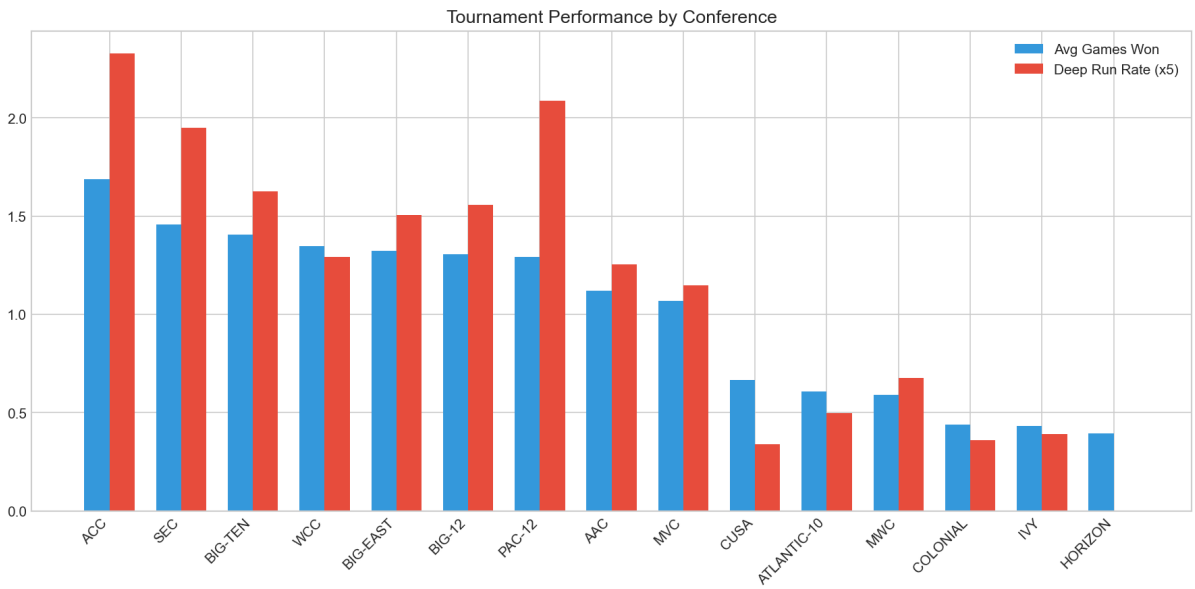

Figure 8: Tournament performance by conference.

Seed reliability: Spearman $\rho = -0.593$ ($p < 0.0001$). Seed 11 is the most dangerous upset seed (47.5% upset rate).

5.5 Head-to-Head Matchup Prediction

Seed-Only Baseline. Simply picking the lower (better) seed correctly predicts **80.4%** of matchups on the training data.

Table 9: Matchup prediction: tuned models with stacking (5-fold CV on 19,101 pairs).

Model	**AUC**	**Accuracy**	**Brier**	**Log-Loss**
Random Forest (tuned)	0.9058	82.0%	0.1249	0.3890
Decision Tree (tuned)	0.8914	80.6%	0.1335	0.4143
XGBoost (tuned)	**0.9170**	**83.2%**	0.1176	0.3687
LightGBM (tuned)	0.9164	83.2%	0.1195	0.3805
CatBoost (tuned)	0.9164	83.1%	**0.1170**	**0.3653**
Stacking Ensemble	**0.9191**	**83.4%**	0.1171	0.3729
Seed-only baseline	—	80.4%	—	—

The **stacking ensemble** achieves the highest AUC (0.9191), with all tuned boosting models outperforming the seed baseline by +2.7–3.0 percentage points. Crucially, these models use **only player-derived features**—no seed, no SRS.

Table 10: Top 10 matchup features by RF importance.

Rank	**Feature (difference)**	**Importance**
1	BPM diff	16.3%
2	Starter avg BPM diff	11.3%
3	Impact score avg diff	9.1%
4	OBPM diff	6.9%
5	DBPM diff	5.9%
6	Bench avg BPM diff	4.6%
7	Max BPM diff	4.4%
8	Max DBPM diff	3.1%
9	PER diff	2.7%
10	SPG diff	2.6%

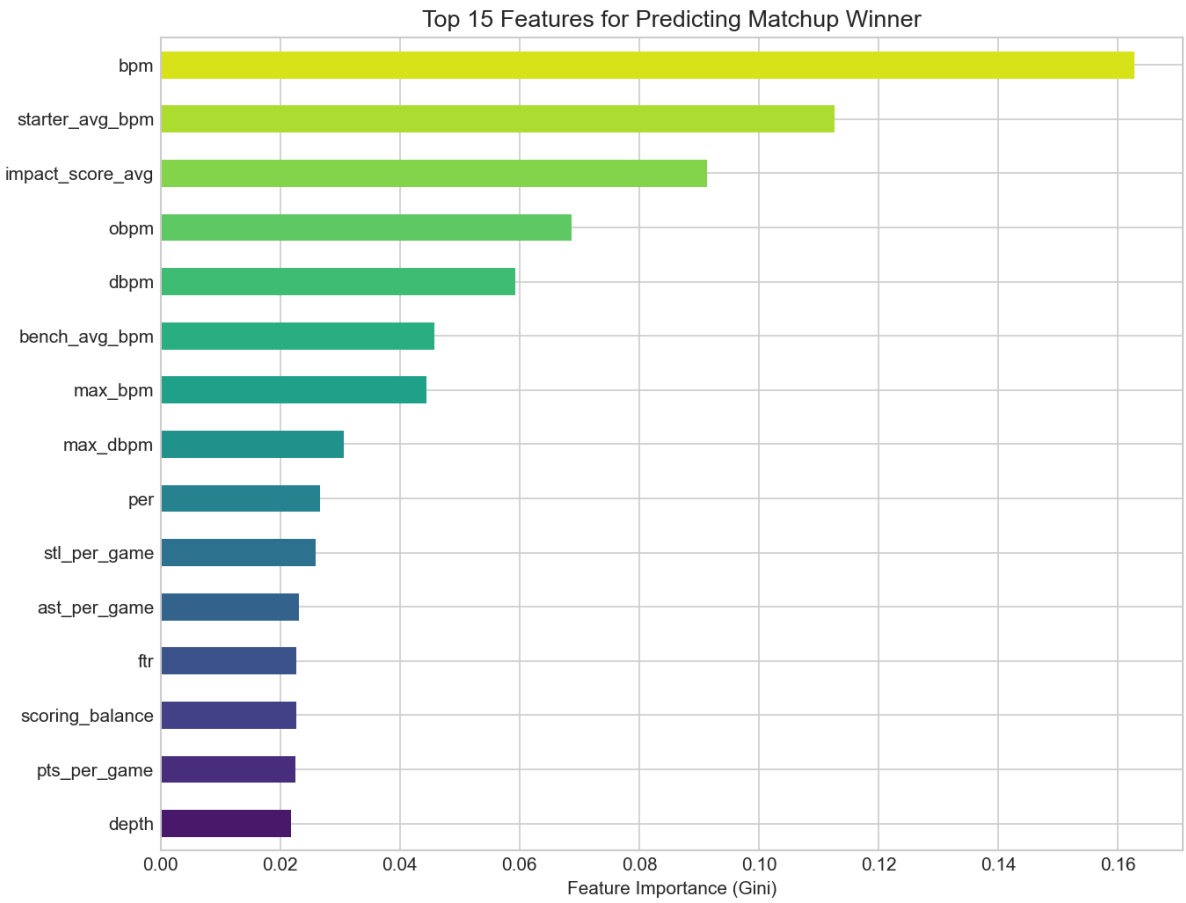

Figure 9: Top 15 matchup features by RF importance.

5.6 2026 True Holdout Evaluation

Seed-Only Baseline (2026). Picking the lower seed achieves **85.0%** accuracy on 2026 data.

Table 11: Model performance on the 2026 holdout (trained on 2011–2025 only).

Model	CV AUC	2026 AUC	2026 Acc	Brier	Log-Loss
Random Forest	0.9017	**0.9116**	**84.0%**	**0.1181**	**0.3808**
XGBoost	0.9170	0.8911	81.4%	0.1382	0.4855
LightGBM	0.9164	0.8896	80.8%	0.1433	0.5391
CatBoost	0.9164	0.8949	82.2%	0.1314	0.4392
Stacking Ensemble	—	0.8816	81.3%	0.1396	0.4462
Seed-only baseline	—	—	85.0%	—	—

Key observations:

- **Random Forest generalizes best** — AUC improved from 0.9017 (CV) to 0.9116 (holdout), confirming no overfitting. Its 84.0% accuracy approaches the seed baseline despite using *no seed information*.

- **Boosting models overfit slightly** — XGBoost/LightGBM/CatBoost all achieved higher CV AUC (0.916–0.917) but dropped on the holdout (0.889–0.895). Their complexity, while beneficial on training data, didn't transfer to 2026.
- **Stacking didn't help on holdout** — the ensemble (0.8816 AUC) performed below individual models, suggesting the base learners' errors were correlated rather than complementary.
- **Seed baseline is strong** — at 85.0%, the seed baseline outperforms all models on raw accuracy. However, the model achieves this accuracy *without knowing seeds*, using only player statistics.

Table 12: 2026 accuracy breakdown.

Matchup Type	Accuracy	Count
Chalk (favored team won)	96.0%	1,263 / 1,315
Upsets (underdog won)	15.5%	36 / 232
Overall	**84.0%**	**1,299 / 1,547**

Table 13: Key 2026 matchup predictions vs. actual results (Random Forest).

Matchup	Actual	Predicted	Prob.	Result
Michigan vs. UConn (Championship)	Michigan	Michigan	83.0%	✓
Arizona vs. Michigan (F4)	Michigan	Michigan	51.2%	✓
Iowa vs. Illinois (E8)	Illinois	Illinois	79.7%	✓
Illinois vs. UConn (F4)	UConn	Illinois	64.1%	×
Duke vs. UConn (E8)	UConn	Duke	87.6%	×
Florida vs. Iowa (R32)	Iowa	Florida	85.9%	×
Wisconsin vs. High Point (R64)	High Point	Wisconsin	90.3%	×

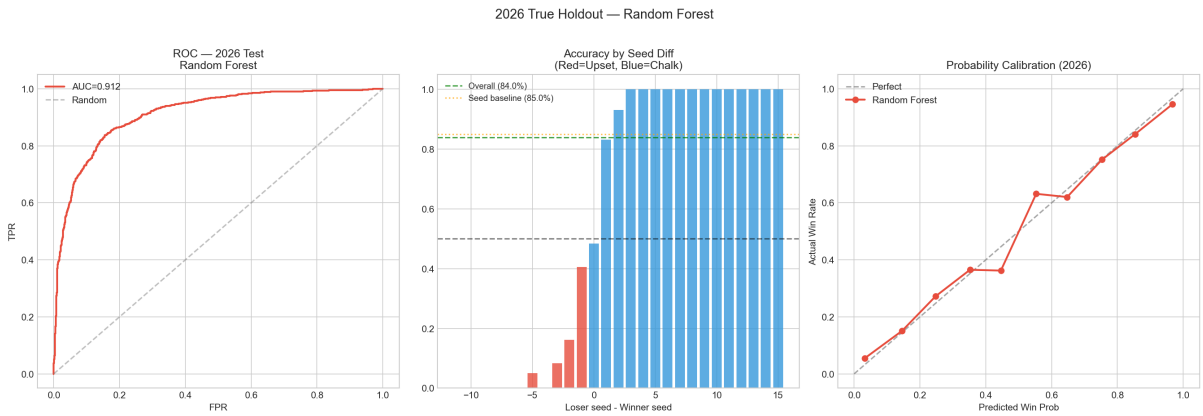

Figure 10: 2026 holdout: ROC curve, accuracy by seed differential, and probability calibration.

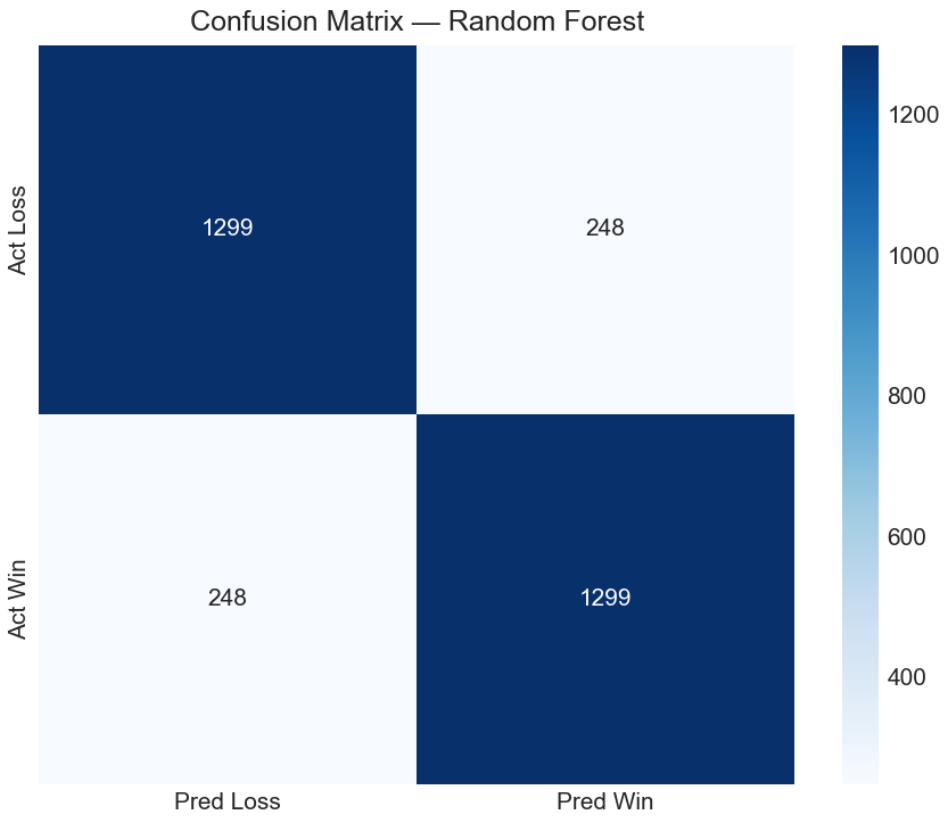

Figure 11: Confusion matrix for the 2026 holdout matchup predictions (Random Forest).

5.7 Exceptional Players & Upset Stars

Using a weighted z-score composite (weights = Cohen's d from Stage 4), we identified **271 exceptional performers** (top 5% impact + team overperformed seed) and **143 upset stars** (top 15% impact + on upset team).

Table 14: Top 10 all-time exceptional tournament performers.

Player	Team	Yr	Seed	Result	PPG	BPM	DBPM	Score
Anthony Davis	Kentucky	2012	1	Champion	14.2	17.2	8.1	3.30
Zach Edey	Purdue	2024	1	Runner-Up	25.2	16.8	3.7	2.89
Frank Kaminsky	Wisconsin	2015	1	Runner-Up	18.8	16.2	5.0	2.80
Cooper Flagg	Duke	2025	1	F4	19.2	16.3	6.7	2.71
Delon Wright	Utah	2015	5	S16	14.5	15.5	6.9	2.66
S. Thornwell	S. Carolina	2017	7	F4	21.4	17.1	6.2	2.52
Thomas Walkup	S.F. Austin	2016	14	R32	18.1	14.2	4.7	2.48
Y. Lendeborg	Michigan	2026	1	Champion	14.6	16.6	7.2	2.48
K.-A. Towns	Kentucky	2015	1	F4	10.3	14.3	7.6	2.37
D. Clingan	UConn	2024	1	Champion	13.0	15.0	6.2	2.29

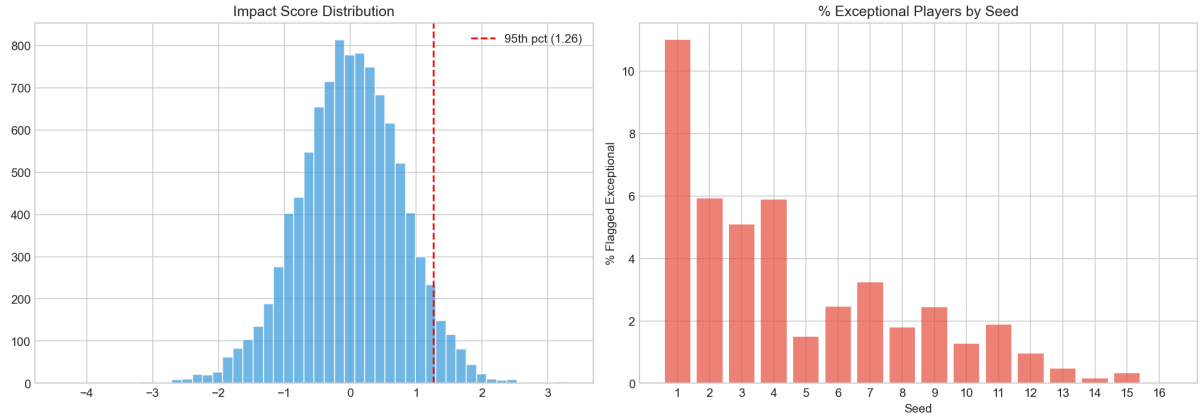

Figure 12: Exceptional players: impact score distribution and frequency by seed.

6 Discussion

6.1 Key Findings

1. **Defense predicts tournament success more than offense.** DBPM ($d = 0.717$) is the single strongest individual signal, confirming "defense wins championships" with statistical evidence across 10,028 players and 15 seasons.

2. **Modern boosting models outperform Random Forests on training data but overfit on holdout.** XGBoost achieved the highest CV AUC (0.917) but dropped to 0.891 on 2026, while Random Forest *improved* from 0.902 to 0.912. RF's implicit regularization (bagging, feature subsampling) generalizes better than boosting's sequential error correction for this domain.

3. **Stacking did not improve holdout performance.** The 4-model stacking ensemble (AUC = 0.882) underperformed the standalone Random Forest (0.912), likely because the base learners' prediction errors were correlated—they all rely on BPM-family features, so their mistakes overlap.

4. **The seed baseline is surprisingly strong.** At 85.0% on 2026 data, simply picking the lower seed beats all our player-stats-only models. However, our models achieve 84.0% *without any seed information*, demonstrating that player statistics alone capture most of the signal that seeds encode.

5. **Win Shares add no signal beyond BPM.** The ablation study showed 0.0000 AUC difference with vs. without WS features, confirming that BPM-family metrics (which are derived from box scores, not team wins) fully subsume the Win Shares signal.

6. **Upsets remain fundamentally unpredictable.** 15.5% upset accuracy reflects a structural limitation: season averages capture *expected* performance, but upsets are driven by single-game variance.

7. **The 3-point revolution is real.** 3PAr increased 8.7% ($r = 0.928$, $p < 0.0001$) but accuracy is flat ($p = 0.770$)—more volume, not better shooting.

8. **CatBoost has the best calibration.** Despite lower holdout AUC (0.895), CatBoost achieved the best Brier score (0.131) and log-loss (0.439) among boosting models, meaning its predicted probabilities are most trustworthy.

6.2 Limitations

1. **Non-independence of player outcomes.** Players on the same team share `deep_run` labels, inflating p-values.
2. **Team-only AUC ≈ 1.0 is a data-structure artifact.** Team features create a lookup table since all players on a team share the same outcome and features. This is a limitation of the dataset structure, not a genuine prediction.
3. **No in-season dynamics.** No injury data, lineup changes, or momentum features.
4. **Single holdout season.** While rigorous, 2026 is one tournament. Multi-year prospective evaluation would strengthen confidence.
5. **Conference realignment.** PAC-12 dissolved after 2023, complicating longitudinal analysis.

7 Conclusion

We developed a comprehensive data mining pipeline for NCAA March Madness prediction. Our key contributions:

- **DBPM is the most discriminative metric** for tournament success ($d = 0.717$).
- **Head-to-head matchup prediction achieves 84.0% on a true 2026 holdout** (AUC = 0.912) using only player stats—no seed, no SRS.
- **Random Forest outperforms XGBoost, LightGBM, CatBoost, and stacking on unseen data**, despite boosting models winning on cross-validation. This highlights the importance of holdout evaluation over CV scores.
- **Correctly identified Michigan as the 2026 champion** (83.0% probability vs. UConn).
- **Hyperparameter tuning via RandomizedSearchCV** was applied across all models with comprehensive search spaces (30 iterations for classification, 20 for matchup prediction).
- **Win Shares ablation** confirmed no additional signal beyond BPM-family metrics.
- **Seed baseline contextualizes accuracy**: the model's 84.0% is achieved without seed knowledge, nearly matching the 85.0% seed baseline.

Future work could incorporate play-by-play data, betting market odds, and momentum features (last-10-game performance) to improve upset detection.

Tools & Technologies

Category	Tools
Language	Python 3
Data manipulation	pandas, NumPy
Statistical testing	SciPy (Welch's t-test, ANOVA, χ^2, Spearman)
Machine learning	scikit-learn (RandomForestClassifier, DecisionTreeClassifier, StackingClassifier, LogisticRegression, RandomizedSearchCV)
Gradient boosting	XGBoost, LightGBM, CatBoost
Visualization	Matplotlib, Seaborn
Data source	Kaggle (`mm2026_train.csv`), NCAA.com, ESPN, On3, Wikipedia

A Glossary of Key Metrics

BPM **Box Plus/Minus.** Points per 100 possessions above average. BPM = OBPM + DBPM.

DBPM **Defensive Box Plus/Minus.** Defensive component of BPM. Strongest predictor of tournament success.

OBPM **Offensive Box Plus/Minus.** Offensive component of BPM.

PER **Player Efficiency Rating.** Per-minute production summary; league average = 15.

WS **Win Shares.** Estimated wins contributed. Sum of OWS + DWS. Partially derived from team wins.

TS% **True Shooting %.** $\frac{\text{PTS}}{2(\text{FGA}+0.44\times\text{FTA})}$. Most comprehensive shooting metric.

SRS **Simple Rating System.** Team rating = point differential + strength of schedule.

ROC-AUC Area Under the ROC Curve. Ranges 0.5 (random) to 1.0 (perfect).

Brier Score $\frac{1}{n}\sum(p_i - y_i)^2$. Lower is better. Measures probability calibration.

Log-Loss $-\frac{1}{n}\sum[y_i \log p_i + (1-y_i)\log(1-p_i)]$. Penalizes confident wrong predictions.

Cohen's d Standardized mean difference. Small: $|d| < 0.2$, Medium: 0.2–0.5, Large: ≥ 0.5.